



TEAM 4 — Attack Path Visualization & Graph Analytics

Introduction

Team 4 is responsible for analyzing how attackers can move through an organization's infrastructure after gaining access to a system. The team focuses on attack path analysis, graph analytics, lateral movement detection, and visualization of relationships between systems, users, vulnerabilities, and services.

The purpose of Team 4 is to transform raw vulnerability and infrastructure data into connected attack graphs that help security teams understand potential attack routes and organizational exposure.

Objective of Team 4

The main objective of Team 4 is to:

- Identify possible attack paths
- Detect lateral movement opportunities
- Analyze privilege escalation routes
- Visualize relationships between systems
- Calculate blast radius impact
- Prioritize high-risk attack chains
- Detect exposure to critical systems

The team helps security analysts understand how attackers may pivot from one compromised system to another inside the network.

Core Responsibilities of Team 4

Team 4 performs several major tasks:

1. Attack Path Analysis

The team identifies possible paths an attacker can use to move across systems.

This includes:

- shortest compromise paths
- multi-stage attack chains
- system-to-system attack relationships
- exposed critical infrastructure

The analysis helps determine how attackers can reach sensitive systems such as:

- domain controllers



- databases
 - cloud environments
 - internal servers
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2. Graph Construction

Team 4 builds a security graph using graph databases such as Neo4j.

The graph contains:

- systems
- users
- services
- vulnerabilities
- credentials
- network relationships

The graph structure allows efficient traversal and attack path discovery.

3. Lateral Movement Detection

Lateral movement analysis identifies how attackers may move from one machine to another after initial compromise.

The system analyzes:

- remote access relationships
- privilege access
- connectivity between hosts
- trust relationships

This helps detect high-risk pivot points within the network.

4. Privilege Escalation Mapping

Team 4 identifies opportunities where attackers may gain higher privileges after compromise.

This includes:

- administrator privilege escalation
- credential abuse
- access chaining
- service privilege exploitation

The goal is to understand how attackers may escalate from normal access to critical administrative control.



5. Blast Radius Analysis

Blast radius analysis determines the potential impact of a compromise.

The system calculates:

- number of affected systems
- reachable infrastructure
- exposed critical assets
- possible compromise spread

This helps organizations estimate the impact of attacks and prioritize remediation.

6. Crown Jewel Exposure Detection

Team 4 identifies whether critical assets are reachable through attack paths.

Critical assets may include:

- domain controllers
- financial systems
- production databases
- cloud management systems

The system determines if attackers can eventually access these sensitive systems.

Workflow of Team 4

Step 1 — Data Collection

Team 4 receives:

- asset information
- monitoring events
- vulnerability data
- infrastructure relationships

This data comes from upstream teams within ThreatGraph.

Step 2 — Data Processing

The collected information is:

- normalized
- enriched
- correlated



- validated

Threat intelligence data may also be added to improve attack analysis.

Step 3 — Graph Generation

The processed data is stored inside a graph database.

Nodes represent:

- systems
- users
- services
- vulnerabilities

Relationships represent:

- connectivity
 - access permissions
 - exploitation paths
 - trust relationships
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Step 4 — Attack Path Computation

Graph algorithms calculate:

- shortest attack paths
- lateral movement routes
- privilege escalation chains
- blast radius exposure

The system continuously updates attack relationships as infrastructure changes.

Step 5 — Visualization

The final attack graphs are visualized using interactive graph dashboards.

Features include:

- attack chain tracing
- graph filtering
- zooming
- risk highlighting
- relationship mapping

These visualizations help analysts understand attacker movement clearly.



Technologies Used by Team 4

Purpose	Technology
Graph Database	Neo4j
Graph Analytics	Neo4j GDS
Backend APIs	FastAPI
Graph Processing	Python
Visualization	D3.js
Interactive Graphs	Cytoscape.js
Streaming	Kafka
Threat Intelligence	OpenCTI, MISP

Importance of Team 4

Team 4 plays a critical role in ThreatGraph because vulnerabilities alone do not show the complete risk.

A single vulnerability may seem low risk individually, but when combined with:

- exposed systems,
- trust relationships,
- credentials,
- and lateral movement paths,

it may lead to complete organizational compromise.

Team 4 helps identify these hidden attack chains before attackers exploit them.

Expected Outcomes

The Team 4 module helps organizations:

- visualize attack paths
 - detect hidden exposure
 - prioritize remediation
 - improve security visibility
 - reduce incident response time
 - understand attacker movement
 - protect critical systems
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Conclusion

Team 4 serves as the attack intelligence and graph analytics layer of ThreatGraph.

By combining graph databases, attack path analysis, and visualization techniques, Team 4 enables organizations to understand how attackers may move through their infrastructure and compromise critical assets.

The module transforms isolated vulnerabilities into connected attack intelligence, helping defenders prioritize remediation and improve overall security posture.

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